
A GROUND-UP QUANTITATIVE FINANCE TEXT

The BGM Model, From First Principles

Rates options, the LIBOR market model, and how
a calibration engine is actually built

DISCOUNT CURVES · CAPS, FLOORS & SWAPTIONS · THE SABR CUBE
HJM & BGM · CASCADE CALIBRATION · MONTE CARLO SIMULATION

Written for a quant or trader building a recalibration application
for rates options, assuming no prior background in the subject

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Preface

This book exists to answer one practical question: *what has to be true, and what has to be built, for a piece of software to take today's cap, floor and swaption prices and hand back a set of model parameters that reprice them?* That software is a BGM — Brace-Gatarek-Musiela — recalibration engine, and it sits on every rates desk that runs exotics. Understanding what it does requires understanding, in order: what a rate actually is and why it behaves unlike a stock price; how the instruments that trade are quoted and priced; where the model came from and why it looks the way it does; how its parameters are fitted to the market; and how the whole thing is wired together into a system that runs every morning before the desk opens.

The book is written so that each of those five questions is answered before the next one is asked. Nothing in Part II assumes you have seen a stochastic differential equation; nothing in Part IV assumes you have calibrated anything. By the time calibration itself is reached in Part V, every object it manipulates — the curve, the caplet, the swaption, the drift — has already been built from scratch and priced by hand at least once.

Who this is for

Written for a quant, quant developer, or trader who needs to build or maintain a BGM calibration application and wants to understand what the code is actually doing, not just which library call to make. No prior background in stochastic calculus, fixed income, or derivatives pricing is assumed; the first three chapters build that foundation from the discount factor upward. A reader who already knows the vanilla rates market can skip to Part III without loss.

By the end, the reader should be able to: price a cap, floor and swaption by hand from a quoted volatility; derive the no-arbitrage drift of a forward rate under an arbitrary measure; explain why a lognormal forward rate model needed inventing at all; calibrate instantaneous volatility and correlation parameters to a live cap and swaption market; and describe, component by component, the architecture of a production recalibration engine.

How to use this book

- **Boxes carry the load.** KEY RESULT boxes hold the formulas worth memorising — if you revise only by flipping through boxes, these are built to be sufficient. PITFALL boxes flag errors that produce plausible-looking wrong numbers, which are far more dangerous than errors that crash a calculation. ON THE DESK boxes explain how the theory is actually used by the people who use it daily.
- **Worked examples are numerical, not symbolic.** Every pricing or calibration formula is followed by a worked example with realistic market parameters, carried through to a sanity check and an interpretation. Several examples — a $1y \times 10y$ payer swaption, a 5-year SOFR cap, a small calibration problem — are reused across chapters so that a single concrete object accumulates meaning as the book progresses.
- **Two practice papers close the book.** Paper A is quantitative — calculation and derivation. Paper B is applied — diagnosing a bad calibration, choosing a correlation structure, deciding what a strange number on a risk report means. Full worked solutions follow both.

- **Each chapter ends with a benchmark.** A short list of things you should be able to do without notes. Treat it as a checkpoint, not a summary.

A caution on sources and conventions

Interest rate markets moved through a genuine regime change during 2021–2023, when LIBOR was retired and replaced by overnight risk-free rates (SOFR in USD, SONIA in GBP, €STR in EUR, among others). This book is written entirely in the post-LIBOR convention: the “forward rate” the model evolves is a forward-looking term rate or a compounded-in-arrears risk-free rate, discounting runs off an OIS/RFR curve, and every numerical example uses realistic 2026 SOFR-market levels. Where a number is indicative of a typical market level rather than a quote observed at a specific instant, this is stated explicitly. Market conventions and typical volatility levels can and do move; before wiring any formula in this book into a system that risks real capital, check it against a primary source and a current market data feed.

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